

RL、RC和RLC电路稳态特性研究2



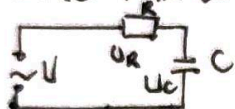
实验(十五).RL,RC和RLC电路稳态特性研究

一. 实验目的

1. 观测 RC, RL, RLC 电路幅频特性相频特性与谐振特性
2. 了解 RC 电路滤波原理与应用
3. 学习用双踪示波器测量信号相位差

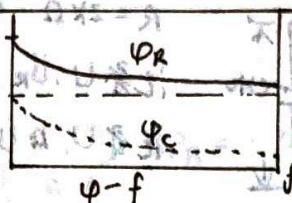
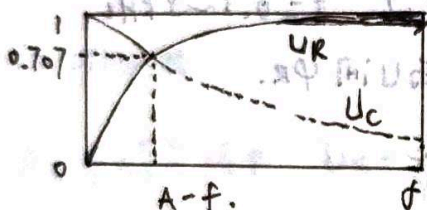
二. 实验原理

1. RC 串联电路.



$$I = \frac{U}{\sqrt{R^2 + (\frac{1}{2\pi fC})^2}} \quad U_R = IR \quad \varphi_R = \arctan \frac{1}{2\pi fCR}$$

$$U_C = \frac{1}{2\pi fC} \quad \varphi_C = -\frac{\pi}{2} + \arctan \frac{1}{2\pi fCR}$$

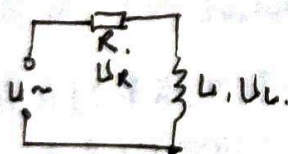


定义 $U_R = U_C$ 时 f 为谐振频率.

$$f_{UR=UC} = \frac{1}{2\pi RC}$$

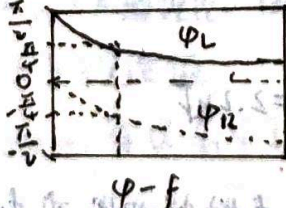
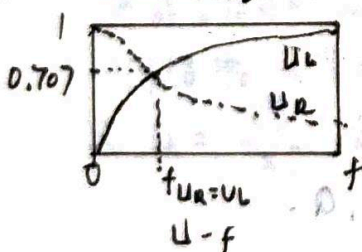
U_R 随 f 增大而增大 可用于高通滤波 U_C 随 f 增大而减小, 可用于低通滤波

2. RL 串联.



$$U_R = \frac{U}{\sqrt{1 + (\frac{2\pi fL}{R})^2}} \quad \varphi_R = \arctan(\frac{2\pi fL}{R}) \quad U_L = \frac{U}{\sqrt{1 + (\frac{R}{2\pi fL})^2}}$$

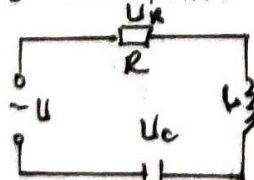
$$U_L \approx \varphi_L = \frac{\pi}{2} - \arctan \frac{2\pi fL}{R}$$



$$f_{UR=UL} = \frac{R}{2\pi L}$$

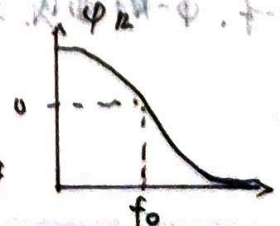
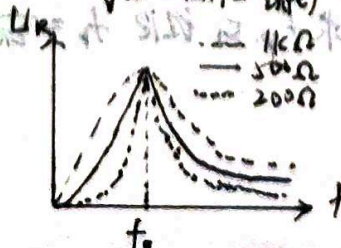
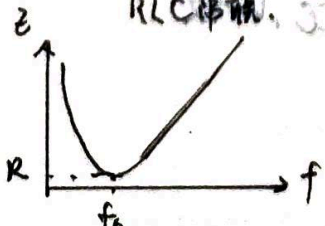
三. 实验主要步骤或操作要点

3. RLC 串联.



$$Z = \sqrt{R^2 + (2\pi fL - \frac{1}{2\pi fC})^2} \quad U_R = \frac{UR}{\sqrt{R^2 + (2\pi fL - \frac{1}{2\pi fC})^2}} \quad U_C = \frac{2\pi fL RU}{\sqrt{R^2 + (2\pi fL - \frac{1}{2\pi fC})^2}}$$

$$U_C = \frac{UR}{\sqrt{R^2 + (2\pi fL - \frac{1}{2\pi fC})^2}} \cdot 2\pi fC \quad \varphi_R = -\arctan \frac{2\pi fL - \frac{1}{2\pi fC}}{R}$$



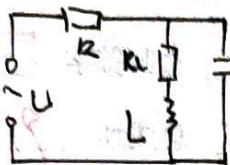
$$f = f_0 = \frac{1}{2\pi\sqrt{LC}} \text{ 时 } U_R = U$$

$\varphi_R = 0$ 等效纯电阻

$$U_L = \frac{2\pi f_0 L}{R} \cdot U = U_C = Q U$$

$$Q = \frac{2\pi f_0 L}{R} = \frac{1}{2\pi f_0 C R}$$

4. RLC 并联



$$Z_p = \sqrt{\frac{R^2 + (2\pi f L)^2}{(1 - 4\pi^2 f^2 L C)^2 + (2\pi f C R)^2}}$$

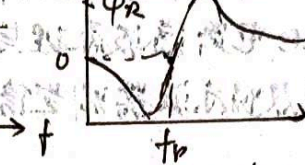
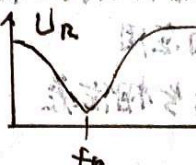
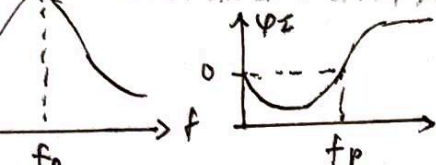
$$\varphi_i = -\arctan \frac{2\pi f L - 2\pi f C(R^2 + (2\pi f L)^2)}{R L}$$

$\varphi_i = 0$ 时呈阻性. 谐振 $f_p = \frac{1}{2\pi \sqrt{LC - (\frac{R}{L})^2}} = f_0 \sqrt{1 - \frac{1}{Q^2}}$ $f_0 = \frac{1}{2\pi \sqrt{LC}}$ $Q = \frac{2\pi f_0 L}{R} = \frac{\sqrt{L/C}}{R}$

$$U_R = U \cdot \sqrt{\frac{R^2(1 + \omega^2 C^2 [C R^2 + L(C L \omega^2 - 1)])}{2 R^2 R + L^2 \omega^2 + R^2 (C L \omega^2 - 1)^2 + R^2 (C^2 R \omega^2 + 1)}}$$

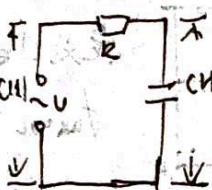
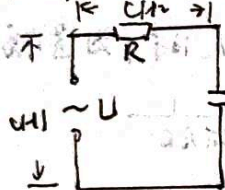
$$\varphi_R = \arctan \frac{\omega L C R^2 + L(C L \omega^2 - 1)}{R L + C^2 R^2 \omega^2 + R(C L \omega^2 - 1)^2}$$

Z_p



并联谐振 Z_p 极大 U_R 极小. 纯阻性. 电感容上 I 为电流与电压 Q 倍

1. 测 RC 串联电路特性.



$R = 2k\Omega$ $C = 47nF$ $f = 0.3 \sim 5kHz$

记录 U, U_R, U_C 与 U 间 φ_R .

记录 U, U_C, φ_C

2. 测 RL

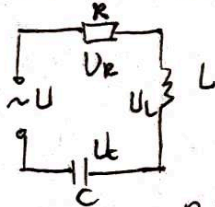


$R = 1k\Omega$ $L = 10mH$ ($R_L \approx 65\Omega$) $f = 0.1 \sim 40kHz$

记录 U, U_R, φ_R . 记录 U, U_L, φ_L

作 $U-f, \varphi-f$ 曲线

3. 测 RLC 串联特性与 Q 值



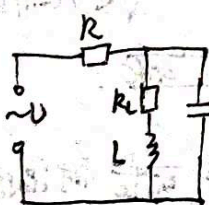
$R = 1000\Omega$ $L = 10mH$ $C = 2.2nF$

记录不同 f 下 U, U_R, φ_R .

作 RLC 电路 $U-f$ 与 $\varphi-f$ 特性曲线求 f_0, Q .

R 取 $500\Omega, 200\Omega$ 测 Q 下 $U-f$ 特性. 作图

4. 测 RLC 并联



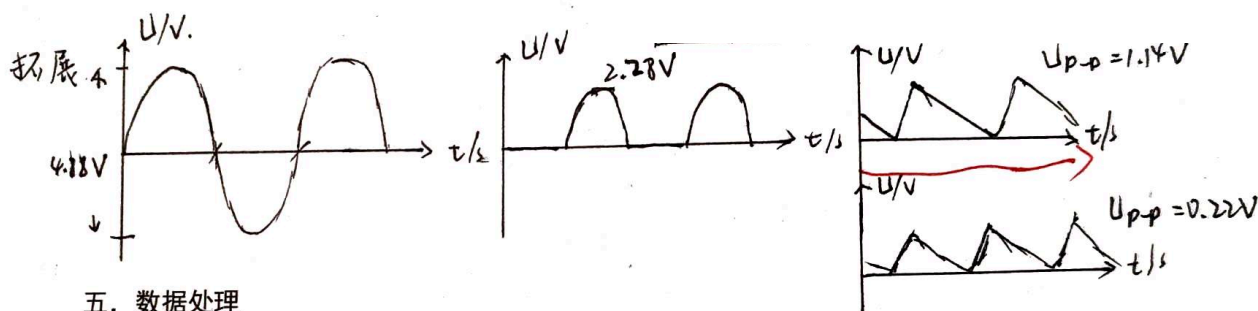
$R = 1000\Omega$ $L = 10mH$ $C = 22nF$ $f = 1 \sim 25kHz$

测 U, U_R, φ_R

作 $U-f, \varphi-f$ 曲线. 求 f_p 与理论 $f_p = \frac{1}{2\pi \sqrt{LC}}$ 比较

四. 实验数据

f/kHz	U/V	U _R /V	φ _R	U _C /V	φ _C						
						f/kHz	U/V	U _R /V	φ _R	U _C /V	φ _C



五. 数据处理

RC串联:

作用有 $f_{UR=UC} \approx 1.70 \text{ kHz}$

$$\text{理论 } f_{UR=UC} = \frac{U}{\sqrt{R^2 + (\frac{1}{2\pi fC})^2}} = I$$

$$\text{则 } f_{UR=UC} = \frac{1}{2\pi RC} = \frac{1}{2\pi \times 2k\Omega \times 47nF} \approx 1.693 \text{ kHz}$$

$$\Delta f = 0.007 \text{ kHz}$$

$$\text{相对误差 } \frac{\Delta f}{f_{UR=UC}} = 0.4\%$$

RLC串, f_0 由图知 $= 34.0 \text{ kHz}$

$$Q_{实验} = \frac{1}{2\pi f_0 C R} = \frac{1}{2\pi \times 34kHz \times 2.2nF \times 1000\Omega} = 2.13$$

$$Q_{理论} = \frac{1}{2\pi f_0 C R}$$

$$f_0 = \frac{1}{2\pi \sqrt{LC}} = \frac{1}{2\pi \sqrt{10mH \times 2.2nF}} = 33.93 \text{ kHz}$$

$$Q_{理论} = 2.132$$

$$\Delta Q = 0.002$$

$$\Delta f = 0.07 \text{ kHz}$$

$$\text{相对误差} = \frac{\Delta Q}{Q_{实验}} = 0.09\%$$

Figure 1 is a graph showing the relationship between the frequency ratio $U_r/U_c f$ (X-axis) and the peak value N (Y-axis). The X-axis ranges from 0.00 to 6.00, and the Y-axis ranges from 0.00 to 6.00. Two curves are plotted: a solid line with circular markers representing U_c/N and a dotted line with circular markers representing U_r/N . The U_c/N curve starts at (0, 0) and increases linearly. The U_r/N curve starts at (0, 0), increases to a peak of approximately 5.5 at $U_r/U_c f \approx 1.5$, and then decreases, crossing the U_c/N curve at $U_r/U_c f \approx 2.5$.

Figure 1 is a graph showing the phase shift (φC/φR-f) versus frequency (kHz) for a series of RC networks. The y-axis is labeled "相位/°" and ranges from -90.0 to 90.0. The x-axis is labeled "频率/kHz" and ranges from 0.00 to 6.00. Two data series are plotted: φR (represented by a dotted line with solid circles) and φC (represented by a solid line with solid circles). Both series show a phase shift that increases with frequency, starting from approximately -10° at 0.00 kHz and reaching approximately 80° at 6.00 kHz. The φC series is consistently higher than the φR series.

Graph showing the normalized voltage $U_r(270\Omega/1k\Omega) - f$ versus frequency f_p (kHz). The graph displays two curves: a solid line for U_r-1k/V and a dotted line for U_r-270/V . The solid line shows a resonance peak around 35 kHz, while the dotted line shows a resonance peak around 35 kHz. The peak voltage for the solid line is approximately 2.1, and for the dotted line is approximately 3.5. A vertical dashed line marks the resonance frequency at 35 kHz. Handwritten red text "35" is next to the peak of the dotted curve.

Figure 1 is a graph showing the phase shift $\phi_r(270\Omega/1k\Omega) - f$ versus frequency f in kHz. The y-axis ranges from -90.00 to 90.00, and the x-axis ranges from 0.0 to 120.0. Two curves are plotted: ϕ_R-1k (solid line with circles) and ϕ_R-270 (dotted line with circles). The curves intersect at $f_p \approx 35$ kHz, where the phase shift is 0. The ϕ_R-1k curve starts at 0 and decreases to -90, while the ϕ_R-270 curve starts at 0 and increases to 90.

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$$W_{R.O} = q \cdot qU$$
$$\sin \alpha = 0.43 =$$

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21.5

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六. 实验结论及现象分析

- 示波器正确连接后, 可观测到信号波形
- 作图得出 $U_R - f$ 与 $U_C - f$ 曲线, 取交点横坐标即为截止频率.
- 得出相对误差 0.4%, 在合理误差范围内, 可以认为测量值 = 理论值
- 作图得 RLC 中 $U_R - f$ 与 $\varphi_R - f$ 曲线, 以 U_R 极值 / $\varphi_R = 0$ 点横坐标为谐振 f .
- 理论与测量误差 0.09%, 在合理范围 $f_0 \approx 34.0 \text{ kHz}$, $Q \approx 2.13$

七. 讨论问题

RL 串联电路中, $U_R = \frac{U}{\sqrt{1 + (\frac{2\pi f C}{R})^2}}$ $\varphi_R = -\arctan \frac{2\pi f C}{R}$

考虑 L 自身电阻后, $R_{实际} > R_{理论}$, 会使 $f_{U_C=U_R_{实际}} > f_{U_C=U_R_{理论}}$
即测得的 $f_{U_C=U_R}$ 增大

在 RLC 串联电路中, $U_R > \dots$ 谐振时, $f_0 = \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}} \cdot \frac{1}{2\pi}$

考虑 L 自身 R' 后, $R_{实际} > R_{理论}$, 由公式知

测得谐振 f 会大于理论值

又 $Q = \frac{\sqrt{L}}{R}$ 从而测得 Q 会小于理论 Q

实验现象观察与原始数据记录

1.

f/kHz	U/V	U_R/V	φ_R	U_C/V	φ_C
0.3	5.36	0.940	79.6°	5.20	-12.4°
0.4	5.36	1.22	77.1°	5.04	-14.0°
0.5	5.36	1.52	70.5°	4.96	-16.6°
0.6	5.36	1.78	68.2°	4.88	-19.9°
1.0	5.36	2.68	55.7°	4.48	-31.8°
1.4	5.36	3.28	49.1°	3.92	-40.9°
1.8	5.36	3.88	39.8°	3.44	-44.7°
2.2	5.36	4.00	36.6°	3.12	-51.1°
2.6	5.20	4.16	33.9°	2.80	-55.4°
3.4	5.20	4.48	26.6°	2.28	-61.4°
4.2	5.20	4.56 4.56	26.4° 23.6°	1.92	-69.1°
5.0	5.20	4.64	18.5°	1.66	-70.0°

2.

f/kHz	U/V	U_R/V	U_C/V	φ_R	φ_C
3.0	5.28	0.240		87.7°	
10.0	5.20	0.820		76.9°	
17.0	5.20	1.60		68.6°	
24.0	5.12	2.72		47.9°	
26.5	5.12	3.16		38.8°	
29.0	4.96	3.52		27.0°	
31.5	4.96	3.76		14.5°	
34.0	4.96	3.84		0.01°	
36.5	4.96	3.76		-11.5°	
39.0	4.96	3.60		-20.9°	
41.5	5.04	3.36		-27.3°	
44.0	5.12	3.20		-34.5°	
72.0	5.20	1.60		-65.1°	
100.0	5.20	1.06		-75.3°	

1kΩ

R L C